

Determining Conversational Styles at Scale

FirstNameA LastNameA ^{1,*}, FirstNameB InitialB LastNameB ^{2,3,*,**}, FirstNameC LastNameC^{1,3}

¹ Address Affiliation 1, Country Affiliation 1

² Address Affiliation 2, Country Affiliation 2

³ Address Affiliation 3, Country Affiliation 3

first@university.edu, second@companyA.com, third@companyB.ai

Abstract

A conversational style is a manner of communication which is often defined along a bipolar axis. Tannen's theory of (spoken) conversational styles is defined on poles of High Considerateness and High Involvement using features like pausing between utterances, loudness and the use of personal topics. In this study on the Switchboard corpus of 138,273 utterances, we use a data-first approach to determine whether the observed data aligns with Tannen's theories. While the effect sizes are small, in most cases, our evidence provides support for Tannen's theory of HC/HI poles; however, we find some minor departures. The results demonstrate that conversational style distinctions can be identified at scale and can be used for future sociolinguistic and computational research.

Index Terms: conversational styles, human-computer interaction, prosody

1. Introduction

Conversational interaction is shaped not only by the content of a dialog but also by patterns in timing, pacing, repetition, and prosody of speakers. These interactional patterns have been defined as conversational styles, prominently in the work of Deborah Tannen, who characterizes conversational behavior along a continuum from High Considerateness (HC) to High Involvement (HI) [1, 2]. The High Involvement style is associated with shorter pauses between utterances, greater utterance length, frequent repetition, faster speech rates, and louder speech volume while the High Considerateness style emphasizes the opposite: longer turn pauses, shorter, quieter, slower, utterances using less repetition.

While these contributions have been highly influential to Natural Language Processing and sociolinguistics, research has mostly focused on qualitative analysis and smaller-scale observational studies [2, 3, 4, 5]. The question remains to be asked whether these speech patterns may be identified at scale in larger speech corpora without predefined labels and thresholds. In this paper, we investigate conversational styles in a large corpus of spontaneous spoken dialogue using an unsupervised, data-driven approach. Utilizing classic unsupervised machine learning techniques such as Principal Component Analysis, this paper investigates whether data organizes into structures representative of the HC/HI continuum. Specifically, this work addresses the following research questions:

1. Is there evidence for Tannen-style High Involvement and High Considerateness conversational patterns in a large speech corpus?

*These authors contributed equally.

**indicates the corresponding author.

2. What interactional and prosodic feature profiles are associated with each conversational style, and are differences between styles statistically significant?
3. How are conversational styles distributed across demographic categories such as generation, gender, region, and education?

By answering these questions, this research provides a model for further identification and exploration of conversation styles at scale. This research produces an annotated version of the Switchboard Speech Corpus that includes features consistent with patterns in HC and HI styles. This resource provides valuable information on demographic, acoustic and prosodic patterns that arise within conversational styles. The findings in the paper help bridge the gap between sociolinguistics and computational speech analysis, which also aids in the development of more efficient conversational agents.

2. Related Work

In her body of work, Deborah Tannen [6, 7, 8] decouples conversational styles from psychology and instead links it to linguistic subcultures based on gender and region. This idea may have had roots decades before [9]. Tannen describes conversational styles as falling on a continuum between High Considerateness (HC) and High Involvement (HI) [4]. High Considerateness is the "softer," more easygoing end of the spectrum, characterized by speech with a relatively slower pace, longer pauses between speaker turns and modest vocal modulation. The opposite pole, High Involvement, conveys more intensity by using more repetitions and questions along with shorter pauses, faster speech rate, or overlapping speech [2].

2.1. Uses of conversational styles

Conversational style analysis has shown many practical uses since its initial conception. Within the last decade, technological and data capturing advancements have enabled the computational study of sociolinguistic theory in new, innovative ways.

One study [10] showed that older people (ages 65 and up) preferred an informal style over formal for a vocal conversational agent when trying to solve tasks, which contrasted with the middle aged group (ages 50-64) who strongly preferred more formality. Another study [11] shows that using phrases with a 2nd person singular ("your heart is a muscle") rather than formal phrasing ("the human heart is a muscle") were useful in a pedagogical setting. While this set of word choices increased the cognitive load for the listener, it also increased retention of the material. Various studies have shown that people prefer to interact with others who match their conversational style [12, 4]. These indications point to the reasonable conclusion that people

have unique conversational agent interaction preferences that can be optimized to enhance their efficiencies. Style matching seems to be widely valued and discussed in modern research, which likely coincides with the recent widespread adoption of commercially-available virtual assistants.

2.2. Limitations with current systems

One problem with chatbots and virtual agents is their limited social interaction abilities. Live speakers phrase their remarks through face management. Face management has the potential to explain comprehensively how speakers will phrase all remarks that have interpersonal implications [13]. Chatbots tend to communicate through a written or spoken language, resulting in a different relationship. Those who speak face to face can side-channel information such as enthusiasm.

Research has demonstrated that personalizing conversational agents to match user-specific preferences can significantly enhance engagement and satisfaction [14, 15]. This work highlights the critical role of customization in overcoming the disconnect often observed in human-computer interactions, such as the uncanny valley. However, this research and other studies similar to it are performed on smaller scale levels due heavily to their costs to conduct, limiting their wide spread applicability.

Recent computational studies have explored the relationship between conversational style and various demographic or situational factors. Research has shown that age, gender, cultural background, and professional context significantly influence conversational behavior [16, 17]. Advancements in models analyzing conversational style based on levels of engagement and relational dynamics have enriched our understanding of how people navigate their interactions [18, 19]. Demographic data and relational dynamics are constantly evolving and must be acknowledged to generalize research findings to wider ranges of people.

Theoretical frameworks like speech act theory and discourse analysis have also contributed valuable insights into conversational mechanics. They explore the performative nature of speech and the role of turn-taking, coherence, and contextual cues in maintaining smooth conversational flow. While initially developed in non-computational settings, these theories remain highly relevant for designing AI systems that seek to replicate human-like dialogue and should be considered when optimizing human computer interaction.

Transformative advancements in natural language processing, particularly with transformer-based models like BERT and GPT, have facilitated significant progress in generating coherent and contextually appropriate text. However, these models often fail to capture the intricate dynamics of rhythm, tone, and cultural cues essential to natural human dialogue. While efforts have been made to integrate sociolinguistic and psycholinguistic insights into these models, fully replicating the complexity of human conversational style remains an open problem [20, 21]. Currently, there seems to be many different ways to connect foundational sociolinguistic theory with modern computational methods and explore their relationship.

3. Data and Features

We use the Switchboard dataset [22], which contains approximately 2,400 telephone conversations (260 hours) from American speakers representing a range of birth years (1924 - 1975), two genders, seven United States regions as well as a “mixed”

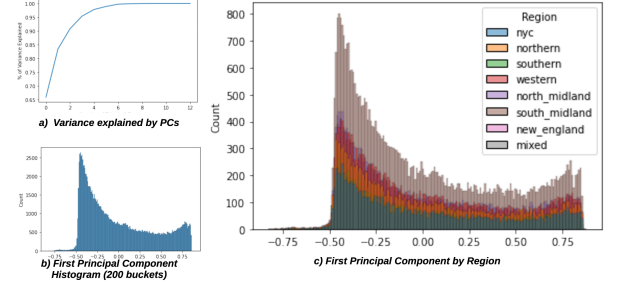


Figure 1: a) Scree Plot of Principal Components. b) Histogram of First Principal Component. c) Histogram of First PC by Region.

eighth region, and various educational levels. Additionally, we determined additional demographics for decade born (eg. 1950s) and Generation (eg. “Baby Boomer”) based on [23].

Table 1: Utterance Features from Switchboard corpus

Class	Feature	Description
Linguistic	Token Count	Token count
Audio	Mean Pitch	Utterance pitch mean
	Pitch StdDev	Utterance pitch std. dev.
	Pitch Range	Utterance pitch range
	Mean Loudness	Utterance loudness mean
	Loudness StdDev	Utterance loudness std. dev.
	Loudness Range	Utterance loudness range
Timing	Turn Gap	Time in seconds between utterances
	Word Rate	Tokens per second
	Syllable Rate	Syllables per second
	Pronoun Rate	Pronouns per second
	Self-repetitions	Rate of token repeats in current utterance
	Prev-repetitions	Rate of token repeats to previous utterance
	Filler Word Rate	Filler tokens (“umm”, “uh”, etc.) per second

Using the timestamped utterance transcripts, we found 138,273 utterances which were not exclusively transcribed as laughter, noise, etc. From each utterance, we extracted the linguistic features from the transcripts, audio (acoustic) features using librosa and timing features from the transcripts, which included timestamps, in Table 1. These features were aligned to the demographic labels for each speaker and assigned a unique utterance name.

4. Method

Since our goal is explainability through facile alignment with sociolinguistic theory, we chose to employ classic machine learning approaches. Using the implementations in scikit-learn [24], we determined principal components on each normalized feature in Table 1, excluding the three pitch-related features: Mean Pitch, Pitch StdDev and Pitch Range. We find these pitch-related features create far less well defined conversational style

groups in the data– see [25] for a richer discussion of human perception of gender given a pitch contour. Exploring the left plot in Figure 1, we find more than 65.91% of variance is explained in the first principal component.

Additionally, we created a histogram of the first Principal Component for the collection of utterances in the middle plot of Figure 1, finding a non-normal distribution.

We examined distributional differences along this first principal component by the demographic labels available through or derived from the corpus, specifically: age demographics (decade, generation), education, gender and region. We find that the general pattern from the middle plot of Figure 1 holds across all demographics. One example, shown in the right plot of the figure, demonstrates the relative similarities across the available region labels, mostly highlighting differences in regional population sizes.

We performed a number of steps, detailed below, on the distribution represented by the general histogram (Figure 1b)):

- We examined the importance of the features which combine to form the first principal component.
- We used a Silhouette analysis to determine an appropriate number of Gaussian components to fit to the distribution, seeking the highest Silhouette score as is practice, varying the number of Gaussian components from 2 to 10. Previewing our results, we find that a mixture of 2 Gaussian components best fits the data.
- We used the Gaussian components resulting from the step above to extract populations (types) of utterances. We examine the feature value differences in those populations.
- Using Welch’s t-test, we investigated whether the feature value differences we find between the utterance populations in the step above are statistically significant. We avoided median-based comparisons because several features (for example, those involved in rates of repetition) frequently take the value of 0. We find that analysis of mean differences provides a more sensitive characterisation of the population-level variation.

5. Results

In this section, we highlight the results of our experiments on Feature Importance, GMM Fitting and its implication and statistical significance of the features found in the different utterance populations.

5.1. Feature importance

We examine the contribution of each feature to the first principal component, which is summarised in Table 2.

The top 3 weighted features (turn gap, token count, and repetitions) directly align to Tannen’s theory as outlined in [2]. The relative weakness of Pronoun rate suggests that speakers from this dataset are not as engaged in the telling of personal stories. This could be attributed to the manner in which the data were collected for this corpus – the speakers are strangers to each other, so may be disinclined toward a more familiar conversational style.

5.2. GMM fitting

We fit a Gaussian Mixture Model (GMM) to the distribution described by the histogram in the middle plot of Figure 1. The results from our Silhouette analysis (left side of Figure 1) indicate that a mixture of 2 Gaussian components best describes the data.

Table 2: Feature importance to Principal Component

Magnitude	Feature
-0.7755	Turn gap
-0.5935	Token count
-0.2039	Self-repetitions
+0.0423	Word rate
-0.0240	Prev-repetitions
+0.0197	Syllable rate
+0.0130	Filler word rate
+0.0131	Loudness std dev
+0.0057	Loudness range
-0.0024	Pronoun rate
+0.0011	Loudness mean

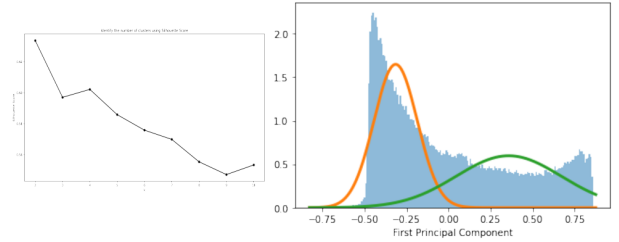


Figure 2: a) Silhouette scores and b) GMM model

We determined the means of each to lie at -0.3145 and 0.3581 of the first principal component. The corresponding standard deviations were estimated as 0.1293 and 0.3119, respectively. The larger standard deviation of the second component reflects greater dispersion along the first principal component, producing a broader, lower peak, while the smaller standard deviation of the first component results in a narrower, taller density concentration. Find these “best fit” Gaussian components on the Switchboard utterances illustrated in the rightmost plot of Figure 2.

Having determined posterior probabilities of association to membership in each Gaussian component, “hard” labels (0 or 1) were assigned using an argmax decision rule over these probabilities. The decision boundary corresponds to the first principal component value at which the posterior probabilities of the two components are equal.

Because of the feature values, we determine that the first Gaussian component – higher peak in Figure 1 – corresponds closely to the High Considerateness (HC) style and the second component corresponds closely to the High Involvement style (HI). We acknowledge that we did not exhaustively investigate each feature associated with Tannen-style conversational styles and that some of her findings are not replicated here. Still, we re-use Tannen’s HC/HI nomenclature in Table 3.

The numbers used in this table are shown to 3 significant places past the decimal point, but they are recorded at higher precision. Each feature (eg. Loudness mean) has differences in values across the two styles, perhaps despite appearance. Results for the differences in population show that the HI style has higher: overlapping speech, tokens and pronouns per utterance, tokens and syllables per second, and both types of repetitions. These results are in line with Tannen’s work. However, contradicting Tannen’s assertions, the HI utterances appear to be relatively quieter. Negative values for turn gaps represent overlapping utterances. Both groups share negative values due to

Table 3: Mean values per Population

Feature	HC Utterances	HI Utterances
Instances in corpus	75785	62488
Year Born (mean)	1953	1951
Education (mean)	2.21	2.35
Turn Gap	-3.897	-3.991
Token count	13.491	13.925
Self-repetitions	3.333	3.689
Word rate	2.423	2.497
Prev-repetitions	2.033	2.151
Syllable rate	3.071	3.151
Filler word rate	0.068	0.063
Loudness std dev	0.011	0.011
Loudness range	0.050	0.049
Pronoun rate	0.132	0.136
Loudness mean	0.009	0.009

the presence of affirmative statements like “ok,” “right,” “hm,” etc. which were left in the utterance transcriptions.

5.3. Comparisons for significance

We evaluated differences between the two GMM derived groups using Welch’s t-tests. All recorded features had statistically significant mean differences ($p < 0.001$). We found that the effect sizes were small (Cohen’s $d \approx 0.02$ – 0.06), indicating subtle but consistent shifts between groups rather than strong separability. Word Rate elicited the largest effect size (.063).

Although medians were also examined, several features (e.g., Repetition Rate) exhibited median values of zero in both groups due to zero-inflation. Consequently, we used mean differences as they provide a more sensitive characterization of group-level variation for these features. Given the findings in Table 3, the result of our tests for significance are listed in Table 4.

Table 4: Significance Tests for Features

Feature	p-value	d-value
Turn Gap	1.52×10^{-06}	-0.026
Token count	3.71×10^{-09}	0.031
Self-repetitions	1.52×10^{-08}	0.052
Word rate	4.19×10^{-31}	0.063
Prev-repetitions	1.32×10^{-06}	0.026
Syllable rate	1.21×10^{-19}	0.049
Filler word rate	1.44×10^{-08}	-0.031
Loudness std dev	3.10×10^{-07}	-0.028
Loudness range	2.06×10^{-13}	-0.040
Pronoun rate	1.88×10^{-08}	0.030

Here, we find that the p-values for all the features are lower than 0.001, indicating statistically significant differences between the two groups. Because the dataset is large, very small differences can yield significant p-values. However, the values observed here are orders of magnitude smaller (10^{-6} to 10^{-31}), suggesting that the effects are robust rather than artifacts of sample size alone. To contextualize the magnitude of these differences, we also report Cohen’s d values. While most features show small effect sizes, several—such as word

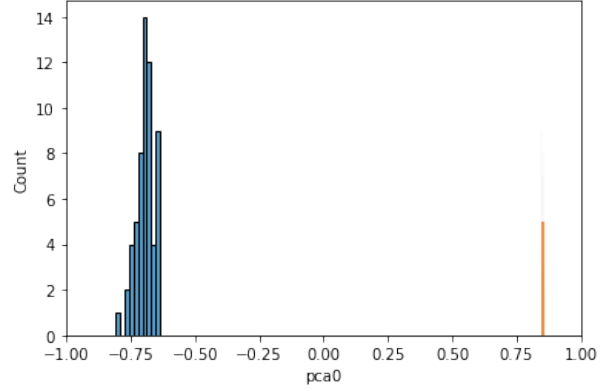


Figure 3: Two SWBD speakers (Conversation ID 4925-B = blue and Conversation ID 2686-B = orange)

rate ($d = 0.063$) and syllable rate ($d = 0.049$)—fall toward the higher end of the observed range, suggesting comparatively strong differences between the two conversational style groups.

6. Discussion

One of the limitations in our study is that we focused on individual utterances as opposed to a speaker’s collection of utterances in one or more conversations. As an illustration: in Figure 3 we see that two speakers (in conversations: 4925 and 2686) exhibit very different patterns. One speaker has utterances exclusively in the HC region while the other shows the opposite. It would be interesting to determine whether individual differences in speaker traits can be attributed to demographics, psychological preferences or other factors. The larger implication is that we did not investigate the behaviour of individuals – for example, whether someone changes their conversational style based on the audience. In that vein, future work would benefit from larger, more contemporary, similarly well-organised corpus with additional coverage and demographic (or other) labels.

7. Conclusion

We took a data-driven approach to finding evidence for conversational styles in Switchboard, a large, well-organised corpus of telephone conversations. We extracted features and used classic machine learning techniques to allow for easy explainability. Having done so, we found evidence of two conversational styles in the data. Our statistically significant findings align closely with Tannen’s theory of a continuum of conversational styles around two poles: High Considerateness (HC) and High Involvement (HI). The HC style has been defined as a slower tempo, leaving more “space” for the speaker’s interlocutor to talk. Our findings also indicate that the HC style is slightly louder and uses more filled pauses (“uh”, “um”, etc.). Some of these deviations from Tannen’s expectations may be due to the participants, the collection methodology or other extraneous factors.

This research provides a model for future exploration and identification of conversation styles at scale which aids in the development of more accessible conversational agents. Future work may add other features, incorporate register to identify other trends, incorporate other diverse corpora, or investigate other conversational styles at scale.

8. References

- [1] H.-W. Dechert, M. Raupach *et al.*, *Psycholinguistic models of production*. Ablex Publishing Corporation Norwood, NJ, 1987.
- [2] D. Tannen, “New york jewish conversational style,” *Intercultural discourse and communication: The essential readings*, pp. 135–149, 2005.
- [3] S. Qiu, U. Gadiraju, and A. Bozzon, “Just the right mood for hit! analyzing the role of worker moods in conversational microtask crowdsourcing,” in *International Conference on Web Engineering*. Springer, 2020, pp. 381–396.
- [4] R. Hoegen, D. Aneja, D. McDuff, and M. Czerwinski, “An end-to-end conversational style matching agent,” in *Proceedings of the 19th ACM International Conference on Intelligent Virtual Agents*, 2019, pp. 111–118.
- [5] I. Tasleem, E. Ramzan, A. Akram, and S. Akram, “Gender differences in indirectness of conversation: A comparative study of male and female communication styles,” *Contemporary Journal of Social Science Review*, vol. 3, no. 3, pp. 1470–1478, 2025.
- [6] D. Tannen, “Spoken/written language and the oral/literate continuum,” in *Annual Meeting of the Berkeley Linguistics Society*, vol. 6, pp. 207–218.
- [7] —, *Gender and discourse*. Oxford University Press, 1994.
- [8] —, ““ don’t just sit there-interrupt!” pacing and pausing in conversational style,” vol. 75, no. 4, pp. 393–395, publisher: Duke University Press.
- [9] G. W. Allport and H. Cantril, “Judging personality from voice,” vol. 5, no. 1, pp. 37–55, publisher: Taylor & Francis.
- [10] J. Chin, S. Desai, S. Lin, and S. Mejia, “Like my aunt dorothy: effects of conversational styles on perceptions, acceptance and metaphorical descriptions of voice assistants during later adulthood,” *Proceedings of the ACM on Human-Computer Interaction*, vol. 8, no. CSCW1, pp. 1–21, 2024.
- [11] L. Lin, P. Ginns, T. Wang, and P. Zhang, “Using a pedagogical agent to deliver conversational style instruction: What benefits can you obtain?” vol. 143, p. 103658, publisher: Elsevier.
- [12] A. Shamekhi, M. Czerwinski, G. Mark, M. Novotny, and G. A. Bennett, “An exploratory study toward the preferred conversational style for compatible virtual agents,” in *International Conference on Intelligent Virtual Agents*. Springer, pp. 40–50.
- [13] T. Holtgraves, “The linguistic realization of face management: Implications for language production and comprehension, person perception, and cross-cultural communication,” pp. 141–159, publisher: JSTOR.
- [14] J. Allen, G. Ferguson, and A. Stent, “An architecture for more realistic conversational systems,” in *Proceedings of the 6th international conference on Intelligent user interfaces*, 2001, pp. 1–8.
- [15] A. B. Kocaballi, E. Sezgin, L. Clark, J. M. Carroll, Y. Huang, J. Huh-Yoo, J. Kim, R. Kocielnik, Y.-C. Lee, L. Mamykina *et al.*, “Design and evaluation challenges of conversational agents in health care and well-being: selective review study,” *Journal of medical Internet research*, vol. 24, no. 11, p. e38525, 2022.
- [16] M. Poser, G. C. Küstermann, N. Tavanapour, and E. A. Bittner, “Design and evaluation of a conversational agent for facilitating idea generation in organizational innovation processes,” *Information Systems Frontiers*, vol. 24, no. 3, pp. 771–796, 2022.
- [17] Y. Rizk, A. Bhandwalder, S. Boag, T. Chakraborti, V. Isahagian, Y. Khazaeni, F. Pollock, and M. Unuvar, “A unified conversational assistant framework for business process automation,” *arXiv preprint arXiv:2001.03543*, 2020.
- [18] S. Gallo, F. Paterno, and A. Malizia, “Conversational interfaces in iot ecosystems: where we are, what is still missing,” in *Proceedings of the 22nd international conference on mobile and ubiquitous multimedia*, 2023, pp. 279–293.
- [19] Y. Gao, Z. Pan, H. Wang, and G. Chen, “Alexa, my love: Analyzing reviews of amazon echo,” *2018 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*, pp. 372–380, 2018.
- [20] R. Kumar and C. P. Rose, “Architecture for building conversational agents that support collaborative learning,” *IEEE Transactions on Learning Technologies*, vol. 4, no. 1, pp. 21–34, 2010.
- [21] Y. Sun and Y. Zhang, “Conversational recommender system,” in *The 41st international acm sigir conference on research & development in information retrieval*, 2018, pp. 235–244.
- [22] J. J. Godfrey, E. C. Holliman, and J. McDaniel, “Switchboard: Telephone speech corpus for research and development,” in *[Proceedings] ICASSP-92: 1992 IEEE International Conference on Acoustics, Speech, and Signal Processing*, vol. 1. IEEE, 1992, pp. 517–520.
- [23] M. Dimock, “Defining generations: Where millennials end and generation z begins,” *Pew research center*, vol. 17, no. 1, pp. 1–7, 2019.
- [24] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, “Scikit-learn: Machine learning in python,” *the Journal of machine Learning research*, vol. 12, pp. 2825–2830, 2011.
- [25] M. Latinus and M. J. Taylor, “Discriminating male and female voices: differentiating pitch and gender,” *Brain topography*, vol. 25, no. 2, pp. 194–204, 2012.